1. Functions in Philosophy of Language and Speech-Acts

1.1. Functions

1.1.1. Seminar 1: Functional explanation and analysis

What is the point of ascribing functions? What theoretical work are they supposed to do? Functional analysis by Cummins (1975)

Functional analysis starts with Hempel and Nagel, sais Cummins. (A) The reason for functional characerization in science is to explain the presence of an item *i* in a system *s*. (B) For something to perform its function is for it to have certain function in *s*. Now we can say that presence of *i* in *s* characterized since *i* is present in *s* ***because*** it has certain effects on *s*. p. 741

Something is wrong, in particular with (A) above.

> An attempt to explain the presence of something by appeal to what it does—its function—is bound to leave unexplained why something else that does the same thing-a functional equivalent-isn't there instead. p. 745

Are functional explanations and teleological explanations of the same kind?

Are functional explanations relative to explanatory interests?

What different kinds of accounts of function are there? Do they have the same explanatory aims (Cummins, 1975; Davies, 2001)?

The short version: both traditions start from Hempel's verdict that functional ascriptions can't explain why an item is present, and they react in opposite ways. Cummins accepts the verdict and changes the question. Wright and Millikan reject it and supply the causal history Hempel thought was missing.

1.1.1.1. Hempel's problem

In "The Logic of Functional Analysis" (1959), Hempel asks whether "the heart beats in order to circulate the blood" can be turned into a deductive-nomological explanation of why organisms have hearts. The natural reconstruction is:

1. System *s* functions adequately in setting *c*. 2. *s* functions adequately only if condition *n* (circulation) is met. 3. If trait *i* (a heart) were present, *n* would be met. 4. Therefore *i* is present.

That is affirming the consequent. To make it valid, premise 3 would have to say that *i* is *necessary* for *n*. But there are functional equivalents: other circulatory arrangements, an artificial heart. So the most you can validly derive is that some member of a class of functional equivalents is present. Functional analysis therefore has heuristic value but no explanatory force for the item itself.

Two features of Hempel's framing matter for what follows: - The explanandum is assumed to be the presence of the item. - Explanation is assumed to mean subsumption of the explanandum under laws.

1.1.1.2. Cummins: functional explanation (causal role)

Cummins ("Functional Analysis", 1975) accepts Hempel's verdict and drops both assumptions.

- **A different explanandum.** Function ascriptions don't explain why the heart

is there. They explain how the containing system has some capacity, such as the circulatory system's capacity to transport nutrients. - **A different form of explanation.** The explanation is *analytic*. You decompose a system's capacity into the capacities of its components and their organization. The function of *x* is whatever *x* contributes to the capacity being analysed, relative to that analysis. This explains a disposition by its constitution, not an event by a law. Cummins contrasts this with the "transition theories" (explaining changes of state) on which the DN model was built. - **Synchronic and history-free.** What the heart did for ancestors is irrelevant. What matters is what it does now in the system under analysis.

The costs follow directly: - **Liberality.** Anything contributing to any capacity we choose to analyse has a function. The heart's function relative to "producing sounds a stethoscope can detect" is to thump. - **No room for malfunction.** A heart that doesn't pump contributes nothing, so on this account it has no pumping function, rather than having one it fails to perform.

1.1.1.3. Wright and Millikan: teleological explanation (etiology)

Wright (1973) and, more rigorously, Millikan (*Language, Thought, and Other Biological Categories*, 1984; "In Defense of Proper Functions", 1989) keep Hempel's explanandum. They hold that a function ascription really does explain why the item is there, and that Hempel's reconstruction failed because it looked at the wrong effects.

- Wright: the function of *X* is *Z* if *X* is there because it does *Z*, and *Z* is a consequence of *X*'s being there. - Millikan's *proper function*: an item has proper function *F* if it is a reproduction (copy) of earlier items that performed *F*, and that performance is part of the causal explanation of why the current item exists. In biology, this means it was selected for *F*.

This dissolves Hempel's problem, because the teleological "in order to" is backed by ordinary forward causation. It is not the current heart's pumping that explains the current heart. It is the pumping of ancestral hearts, through differential reproduction. Functional equivalents also stop mattering. We are not deducing the heart from the need for circulation. We are giving the actual causal history, and an artificial pump that could have done the job was never part of it.

The approach also earns normativity. A defective heart still has the proper function of pumping, because the function is fixed by its lineage, not its performance. That is why Millikan can use proper functions to ground representation and meaning, where misrepresentation needs the same structure as malfunction. The well-known costs: - Swampman: an atom-for-atom duplicate with no history has no proper functions. - Function ascriptions become hostage to evolutionary facts we often don't know.

Millikan's answer to the second is that she is giving a theoretical definition, not an analysis of what speakers mean by "function".

1.1.1.4. How each breaks with Hempel

	Cummins	**Wright / Millikan**
Explanandum	a capacity of the containing system	the presence of the trait
Direction	synchronic, part-to-whole	diachronic, historical-causal
Form	analysis / decomposition, not laws	selection history, a causal narrative
Hempel's verdict	accepted: functions don't explain presence, but they explain something else	rejected: they explain presence, through past tokens' effects
Normativity	none	yes (malfunction possible)

Both abandon the idea that explanation must be DN subsumption. Cummins does so openly, since capacity analysis isn't nomological at all. Millikan's selection explanations are causal but historical and token-lineage based, not derivations from general laws.

The terminology isn't fixed. Some authors call both "functional" and reserve "teleological" for the etiological view; others (Godfrey-Smith's "modern history" theory, Kitcher's design-based unification) try to bridge them. The working rule is: Cummins = *what it does in the system*; Millikan = *what it was selected for*.

The split maps neatly onto your old field. Ecosystem "function" (nutrient cycling, a species' role in a food web) is Cummins-style. Ecosystems aren't units of selection, so there are no etiological functions at that level. Organism-level traits get both kinds of ascription. That is one reason the "ecosystem function" debate in philosophy of ecology keeps turning on exactly this distinction.

1.1.2. Seminar 2

What are 'direct proper functions'? What are they supposed to explain?

> A function F is a direct proper function of x if x exists having a character C because by having C it can perform F. (Notice how close this is to the idea that x exists in order to perform F.) First interpret 'because by having C it can perform F' to mean 'because there were things that performed F in the past due to having C.' But how could it be because F was performed in the past by C as opposed to, merely, because F was performed in the past by something or other, that x was produced? Consider a motor idling. A cyclical process is involved here.

$$Cx \wedge [\ \exists y\ (A(y,x) \wedge Cy \wedge (Cy \ \triangleright\ Fy))\ \triangleright\ Cx\ ]$$

What are 'Normal explanations'?

What are derived proper functions and what is their role in the theory?

What is the difference between the Generalized Selected Effects theory of function and Millikan's?

Are etiological theories of function too liberal in ascribing functions (Millikan, 1984; Garson, 2017)?

1.1.3. Seminar 3

What is the indeterminacy problem?

> While teleosemantics is widely regarded as one of the most attractive approaches to the naturalization of intentionality, many, perhaps all, teleosemantic theories on offer

> are subject to an embarrassing indeterminacy problem: they are not able to make univocal predictions as regards the content of mental states.

But why should we expect that the content of our minds should be possible to explain teleologically?

Why is the possibility of malfunctioning relevant for functional classification? Is an etiological approach necessary for explaining this possibility?

How do 'by'-relations help solve indeterminacy problems?

Why are vestigial traits a problem for etiological approaches?

Can etiological functions explain properly explain malfunctioning? Can systemic functions do so (Neander, 1995; Davies, 2001)?

1.1.4. Seminar 4

Can the notion of function be thought of in terms of 'design'?

What is the difference between an 'initial presence' and a 'maintenance' view of etiological functions?

Does 'design' provide a way to unify other perspectives on function?

What distinguishes a 'recent history' view?

Are etiological functions and system functions notions which figure in different kinds of explanations?

Should we expect or aim for a unified theory of functions or accept pluralism about the notion (Kitcher, 1993; Godfrey-Smith, 1994)?

1.1.5. Seminar 5

[In Millikan, the discussion on Gricean intentions is not of main importance for the course. Neither are the appendices of Ch. 4. For the Martinez paper, focus on the ideas of the argument and don't worry too much about the formalisms.]

What are stabilizing functions of language devices?

How do they relate to the 'literal use' of language devices?

What is the connection between a speaker's purpose of utterance and its function?

How do functions relate to meaning on Millikan's view?

How do teleosemantic approaches aim to provide a basis for representational views of meaning? Why might one be sceptical of such attempts (Millikan, 1984; Martinez, 2013)?

1.1.6. Seminar 6

What sense of function is Sellars appealing to?

How does Sellars view lend itself to avoiding

What is an EMU and what theoretical work are they supposed to do?

What sense of function is Williams discussing?

Can inferential role be considered functionally?

Is Williams sketching a view of how to relate two different senses of 'linguistic function' to each other (Sellars, 1973; Williams, 2013)?

1.1.7. Seminar 7

What notions of function are Cappelen willing to entertain? How do they relate to the approaches we've looked at so far?

Why does Cappelen think that 'function' is a flawed notion for investigating language?

How does Thomasson answer that criticism? Is she successful?

How should we categorise Thomasson's notion of function?

Are functional considerations relevant for conceptual engineering/ethics? In which ways (Cappelen, 2018; Thomasson, 2020)?

1.1.8. Seminar 8

What are the intuitions which support conventionalism and intentionalism about speech-acts?

Why might one be sceptical about the possibility that all speech-acts can be characterised in terms of conventions?

What resources are available to a conventionalist for responding to such criticism?

Which approach seems more promising (Harris, D. W. *et al.*, 2018; Strawson, 1964; Harris, Daniel W., 2016)?

1.1.9. Seminar 9:

How does Millikan analyse conventions? Does it relate to her views on function?

How does Millikan propose to bridge the gap between conventionalism and intentionalism?

What characterises 'co-operative proper functions'? Why does Millikan think that these characterise speech-acts?

How does communicative function relate to conversational score?

Are Murray and Starr using the same notion of function as Millikan? Does this generate a distinct kind of functionalism (Millikan, 2005; Murray and Starr, 2018)?

1.1.10. Seminar 10

Each of the readings suggests an analysis of a particular speech-act in terms of its function. In what sense of function are they best understood?

Is it plausible that these speech-acts have only a single function? If not, how does that interact with a functionalist approach to characterising speech-acts?

References

Cappelen, H. in *Fixing Language: An Essay on Conceptual Engineering,* Oxford University Press, Oxford (2018). https://doi.org/10.1093/oso/9780198814719.001.0001.

Cummins, R., "Functional Analysis," *The Journal of Philosophy* **72**, pp. 741–760 (1975). https://doi.org/10.2307/2024640.

Davies, P. S., *Norms of nature: Naturalism and the nature of functions,* MIT Press, Cambridge (2001). https://doi.org/10.7551/mitpress/5043.001.0001.

Garson, J., "A Generalized Selected Effects Theory of Function," *Philosophy of Science* **84**, pp. 523–543 (2017). https://doi.org/10.1086/692146.

Godfrey-Smith, P., "A Modern History Theory of Functions," *Noûs* **28**, pp. 344–362 (1994). https://doi.org/10.2307/2216063.

Harris, D. W., Fogal, D. and Moss, M., "Speech Acts: The Contemporary Theoretical Landscape" in *New Work on Speech Acts,* ed. Fogal, D., D. W. Harris and M. Moss, pp. 1–39, Oxford University Press, Oxford (2018). https://doi.org/10.1093/oso/9780198738831.003.0001.

Harris, Daniel W., "Intentionalism versus The New Conventionalism," *Croatian Journal of Philosophy* **16**, pp. 173–201 (2016).

Kitcher, P., "Function and Design," *Midwest Studies in Philosophy* **18**, pp. 379–397 (1993). https://doi.org/10.1111/j.1475-4975.1993.tb00274.x.

Martinez, M., "Teleosemantics and Indeterminacy," *Dialectica* **67**, pp. 427–453 (2013). https://doi.org/10.1111/1746-8361.12039.

Millikan, R. G. in *Language, Thought, and Other Biological Categories,* Bradford Books/MIT Press, Cambridge MA (1984). https://doi.org/10.7551/mitpress/4124.001.0001.

Millikan, R. G., "Language Conventions Made Simple" in *Language: A Biological Model,* ed. Millikan, R. G., pp. 1–23, Oxford University Press, Oxford (2005). https://doi.org/10.1093/0199284768.003.0001.

Murray, S. E. and Starr, W. B., "Force and Conversational States" in *New Work on Speech Acts,* ed. Fogal, D., D. W. Harris and M. Moss, pp. 202–236, Oxford University Press, Oxford (2018). https://doi.org/10.1093/oso/9780198738831.003.0009.

Neander, K., "Misrepresenting and Malfunctioning," *Philosophical Studies* **79**, pp. 109–141 (1995). https://doi.org/10.1007/BF00989706.

Sellars, W., "Meaning as Functional Classification," *Synthese* **27**, pp. 417–437 (1973). https://doi.org/10.1007/BF00484606.

Strawson, P.F., "Intention and Convention in Speech Acts," *The Philosophical Review* **73**, pp. 439–460 (1964). https://doi.org/10.2307/2183301.

Thomasson, A. L., "A Pragmatic Method for Normative Conceptual Work" in *Conceptual Engineering and Conceptual Ethics,* ed. Burgess, A., H. Cappelen and D. Plunkett, pp. 435–458, Oxford University Press, Oxford (2020). https://doi.org/10.1093/oso/9780198801856.003.0021.

Williams, M., "How Pragmatists Can Be Local Expressivists" in *Expressivism, Pragmatism, and Representationalism,* pp. 128–144, Cambridge University Press, Cambridge (2013). https://doi.org/10.1017/CBO9780511842498.009.